SUGGESTIVE QUESTIONS OF ICSE BIOLOGY 2026 (13-03-2026 - Friday)

CHAPTERS TO BE LEARN WITH SPECIAL EMPHASIS
CHEMICAL CO ORDINATION IN PLANTS & EVOLUTION
(Chapters to be given thorough emphasis and nothing to be left from these two chapters)

1. Reproductive System with special emphasis on Human Male and Female reproductive system and also Post fertilization Mechanism along with Foetus diagram. Foetus inside the Uterus is most important along with the Diagram of Fertilization and logical sequential stages of Fertilization
2. Cell Division with special reference to Mitosis (Late prophase, Metaphase, Early Anaphase and Late Anaphase and Telophase) and difference between Mitosis and Meiosis.
3. Fundamentals of Genetics with special reference to Monohybrid Cross, Dihybrid Cross, Sex linked inheritance and examples of Sex-linked inheritance along with the numericals.
4. Absorption by Roots.
5. Transpiration.
6. Photosynthesis.
7. Chemical Coordination in Plants with special reference to Phyto hormones, their physiological effects and rich regions in the plant where they are available.
8. Circulatory System with special emphasis on Cardiac Cycle, Structure of Heart and Blood Vessels particularly T.S. of Artery, Vein and Capillary.
9. Excretory System with special reference to Malpighian Corpuscles.
10. Nervous System with special emphasis on Brain, Ear & Eye. (Brain & Reflex Action is highly important).
11. Sense organs, both the diagrammatic representation of Eye and Ear with special reference to Membranous labyrinth, Myopia and Hypermetropia. Accommodation of Eye as well as Light and Dark Adaptation.
12. Endocrine System. (Chart format)
13. Evolution with special reference to the comparative chart on human evolution, Also elongation of giraffe's neck from the perspective of Lamarckian & Darwinian Concept. Industrial melanism with special reference to Biston betullaria.
14. Human Population with special reference to Population Explosion—its Definition, effects, consequences and remedial measures with the concept of Tubectomy and Vasectomy.
15. Environmental Pollution with Abatement of each type of Pollution. Most important for ICSE 2026 are: - AIR, WATER, NOISE, SOIL & Plastic pollution. Also, Swachh Bharat Abhiyan and its objectives. Expanded form of ODF is Open Defecation Free and DDT as Dichlorodiphenyl Trichloroethane.

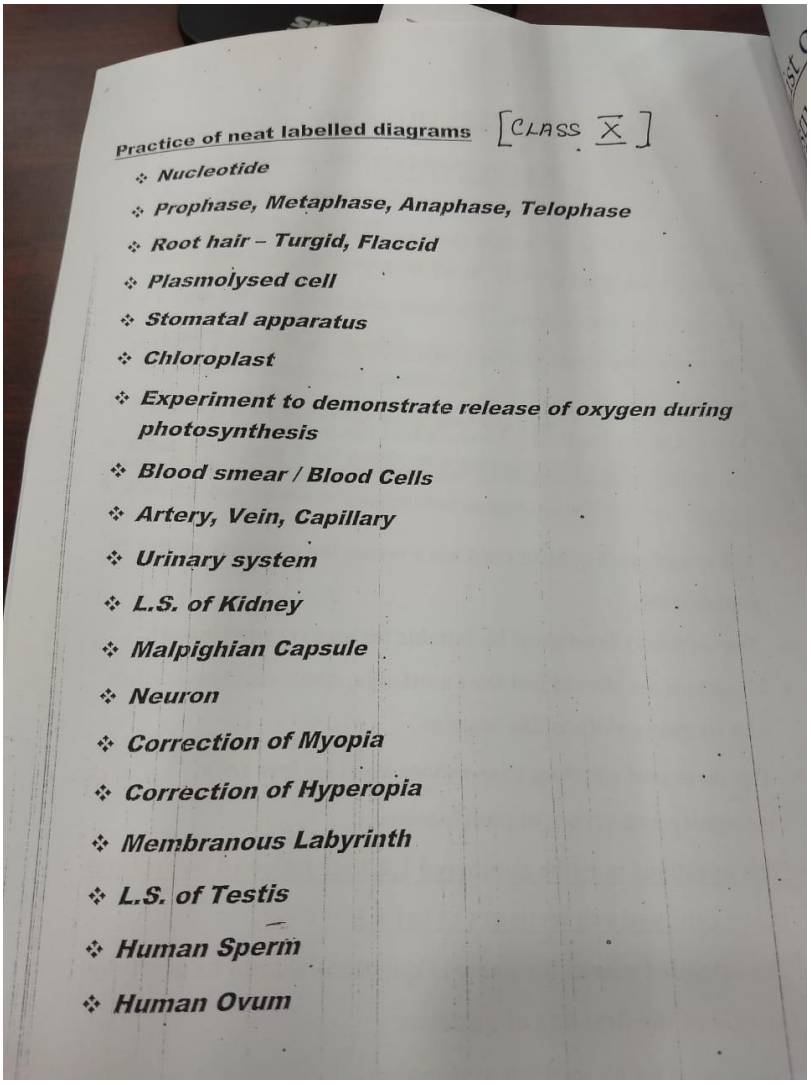
DIAGRAMMATIC QUESTIONS REQUIRES FOR LABELLING

1. Mitotic stages with special reference to Late Prophase, Metaphase, Early Anaphase, And Late Anaphase. (Important question for 2026)
2. Structure of a double helical DNA specifying Hydrogen bonds and Nucleotides. (Important question for 2026)
3. To demonstrate that roots absorb water/ Transpiration Pull in Plants. (Important question for 2026)
4. To demonstrate Endosmosis in plants by Cellophane paper/ Visking bag/ Potato Osmoscope.
5. Demonstration of Plasmolysis and Deplasmolysis. (Important question for 2026)
6. Girdling Experiment. (To demonstrate the role of Xylem & Phloem). (Important question for 2026)
7. Demonstration on Transpiration with polythene bag and Bell jar. (Important question for 2026)
8. Ganongs Potometer. (Important question for 2026)
9. To demonstrate that more number of stomata is present in the abaxial surface (lower surface) of a dicot leaf. (Important question for 2026)

10. To demonstrate that Oxygen gas bubbles is given out during Photosynthesis. (Important question for 2026)
11. To demonstrate that CO_2 is necessary for Photosynthesis. (Moll's half leaf Expt.) (Important question for 2026)
12. Effect of Auxin and Gibberrelins on the Plant growth with reference to sunlight (Important question for 2026)
13. Different types of tropic movements with special reference to Thigmotropism, Phototropism, Geotropism and Chemotropism. (Important question for 2026).
14. Clinostat along with its important role in tropism. (Important question for 2026)
15. T.S. of Artery, Vein and Capillary, identification of various parts. (Important question for 2026)
16. Diapedesis & Phagocytosis of Leucocyte (WBCs). (Important question for 2026)
17. Structure of Human Heart. (Important question for 2026)
18. Cardiac Cycle with special reference with Ventricular Systole & Auricular Systole. (Important question for 2026)
19. Double Circulation of Human Heart with all of the names of Blood vessels. (Important question for 2026)
20. Identification of blood cells in a blood smear. (Important question for 2026)
21. Excretory System of Human. (Important question for 2026)
22. Uriniferous tubule (Nephron) with its blood vessels with special reference to Renal Artery, Afferent Arteriole, Glomerulus, Bowman's capsule, Efferent Arteriole and Renal Vein. (Important question for 2026)
23. V.S. of Human Brain. (Important question for 2026)
24. V.S. of Human Ear and Membranous Labyrinth (Internal Ear) (Important question for 2026)
25. Diagram for V.S. of Human Eye. (Important question for 2026)
26. Reflex Arc. (Important question for 2026)
27. Defects of Human Eye with special reference to Myopia and Hypermetropia. (Important question for 2026) specially Hypermetropia but don't leave Myopia.
28. Male Urinogenital System labelling. Labelled Diagram of Spermatozoa. (Important question for 2026)
29. Female Reproductive System (frontal view). (Important question for 2026)
30. Human foetus in the womb. (Important question for 2026)
31. Stages and sequences of Human fertilization. (Important question for 2026)
32. Overall endocrine glands of humans, secretion of Hormones their functions, Effect on human body under the situation of Hypo secretion and Hyper secretion. (Important question for 2026)
33. Pictorial representation and Diagram based questions on Adrenal, Thyroid, Pituitary and Pancreas. (Important question for ICSE 2026)
34. Role of Pancreatic hormone (Insulin and Glucagon) in regulating blood glucose level. (Important question for 2026)
35. Vasectomy and Tubectomy with special reference to different parts of Male and Female reproductive structures. (Important question for 2026)
36. Industrial melanism with reference to *Biston betullaria* (Important question for 2026)
37. Different pictorial representation of different stages of Human Evolution (Important question for 2026)
38. Elongation of giraffe's neck from Lamarckian and Darwinian viewpoint (Important question for 2026)
39. Water & Plastic Pollution (Important question for 2026)
40. Noise Pollution & Air Pollution (Most important among all types of pollution) (Important question for 2026)
41. Swachh Bharat Aviyan (Important question for 2026)

DIAGRAM NEEDS TO BE PRACTICE FOR DRAWING.

1. All the stages of Mitotic Cell Division (IPMAT) along with Cytokinesis of both Plant with Phragmoplast /Cell Plate formation and Animal with Cleavage Furrow. (Important question for 2026)
2. Stomata with Epidermal Cells both in Open condition & closed condition. (Important question for 2026)
3. Ultra structure of Chloroplast. (Important question for 2026)
4. Human Blood Smear. (Important question for 2026)
5. Typical Neuron. (Important question for 2026)
6. Membranous Labyrinth. (Important question for 2026)
7. V.S. of Human Kidney. (Important question for 2026)
8. V.S. of Human Eye. (Important question for 2026)
9. Defects of Human Eye with special reference to Myopia and Hypermetropia and its rectification with concerned correct lenses. (Important question for 2026)
10. L.S. of Spinal Cord. / Reflex Action and reflex arc. (Important question for 2026)
11. Male Urinogenital System. / L.S. OF Human Testis
12. Female Reproductive System (Only Front View) (Important question for 2026)
13. Human Sperm. (Important question for 2026)
14. Stages of Fertilization. (Important question for 2026)



LIST OF DIAGRAMS

- 1. Duplicated chromosome
- 2. Basic structure of a nucleotide
- 3. Stages of Mitosis
- 4. A single fully grown root hair
- 5. Plasmolyzed root hair
- 6. Plasmolyzed cell
- 7. Ganong's Potometer
- 8. Open/closed stomata
- 9. Chloroplast
- 10. Experiment to show that oxygen is evolved during photosynthesis

LIST OF DIAGRAMS

- 11. Cross section of Artery, Vein and Capillary
- 12. Heart
- 13. The human urinary system
- 14. L.S of kidney
- 15. Malpighian capsule and tubule.
- 16. Neuron
- 17. Brain
- 18. The reflex arc
- 19. External and internal structure of eye.
- 20. Short sightedness and Long sightedness and its correction

LIST OF DIAGRAMS

- 21. Ear
- 22. Membranous Labyrinth
- 23. Thyroid gland
- 24. Adrenal gland
- 25. Male reproductive system
- 26. L.S of testis
- 27. Female reproductive system
- 28. Sperm/ male gamete
- 29.Ovum / female gamete
- 30. Tubectomy & Vasectomy

TOPICS.

BASIC BIOLOGY

1. Location and functions of various cell organelles with special reference to Chloroplast, Centrosome, Mitochondria, Ribosome, Lysosome & Golgi complex (Only chart).
2. Different types of cell division--- Mitosis, Amitosis & Meiosis with their significance of each type. (Important question for ICSE 2026)
3. Synapsis ,Crossing over and Chiasmata:- Definitions only. (Important question for ICSE 2026)
4. Differences between Mitosis & Meiosis. (Important question for ICSE 2026)
5. Types of Chromosomes on the basis of centromere. (Important question for ICSE 2026)
6. Karyokinesis & Cytokinesis. (Important question for ICSE 2026)
7. Morphology of Chromosome & DNA (Important question for ICSE 2026)
8. Concept of Nucleosome, Nucleoside & Nucleotide. (Important question for ICSE 2026)
9. Double helical structure of DNA with Hydrogen bond proposed by Watson & Crick. (Important question for ICSE 2026)
10. Basic terminologies involved in the Fundamentals of Genetics with special reference to Gene, Allele, Homozygous, Heterozygous, Phenotype, Genotype & Mutation. (Important question for ICSE 2026)
11. Mendelian laws of Inheritance with special reference to Law of Dominance and Segregation (Mono hybrid Cross.) & Law of Independent Assortment (Dihybrid Cross) (Important question for ICSE 2026)
12. Mendel experiment of Monohybrid & Dihybrid Cross experiment (Important question for ICSE 2026)
13. Analytical problems relating to Sex Determination & Sex linked Inheritance with special reference to Haemophilia & Colour blindness. (Important question for ICSE 2026)
    NO PEDIGREE ANALYTICAL PROBLEMS ARE EXPECTED IN ICSE 2026

PLANT PHYSIOLOGY

14. Concept of Absorption, Imbibition, Diffusion, Endosmosis, Exosmosis, Plasmolysis & Deplasmolysis with special reference to their DIAGRAMS. (Important question for ICSE 2026)
15. Concept of Hypotonic, Hypertonic & Isotonic Solution. (Important question for ICSE 2026)
16. Concept of Turgidity & Flaccidity.
17. Concept of Wall Pressure, Turgor Pressure, Root Pressure & Osmotic Pressure. (Important question for ICSE 2026)
18. Definition & Types of Transpiration. (Important question for ICSE 2026)
19. Mechanism of Stomatal transpiration & Opening and Closing of stomata. (Important question for ICSE 2026)
20. Transpiration is a necessary evil. Importance and demerits of transpiration. (Important question for ICSE 2026)
21. Methods to reduce excessive transpiration. (Important question for 2026)
22. Factors affecting the rate of transpiration (Both external & Internal factors) (Important question for ICSE 2026)
23. Concept of Anti transpirants with examples.
24. Ultra structure of Chloroplast with special reference to Grana & Stroma. (Important question for ICSE 2026)
25. Definition of Photosynthesis along with the Balanced Chemical Equation (Important question for 2026)
26. Mechanism of Light Phase / Photochemical phase of Photosynthesis. (Important question for ICSE 2026)
27. Overall Balanced Chemical equation for Dark Phase + Polymerization Definition (Important question for ICSE 2026)
28. Concept of Photophosphorylation. (Important question for 2026)
29. Factors affecting the rate of Photosynthesis. (Important diagram-based question for ICSE 2026)
30. Significance of Photosynthesis.
31. Full Chapter on Chemical Coordination in Plants (Important question for 2026)

32. Different components of Human Blood along with their functions. (Important question for ICSE 2026)
33. Differences between Erythrocytes & Leucocytes, Blood & Lymph. (Important question for 2026)
34. External covering & Internal Structure of Human Heart. (Important question for ICSE 2026)
35. Cardiac Cycle & it's different phases (LUBB & DUPP) (Important question for 2026)
36. Concept of SAN & AVN. (Important question for ICSE 2026)
37. Concept of Blood Pressure & Sphygmomanometer.
38. Concept of Blood Vessels- Artery, Vein & Capillaries. (Important question for 2026)
39. Structure of Nephron & Mechanism of Urine formation along with the three steps ------glomerular Ultrafiltration, selective reabsorption & tubular secretion. (Important question for 2026)
40. Normal and abnormal components of Human Urine. (Important question for ICSE 2026)
41. Generalized Structure of Neuron along with its diagram. (Important question for ICSE 2026)
42. Definition of Synapse & Neurotransmitter (Acetylcholine). (Important question for 2026)
43. Classification of Nervous system with conceptual approach of Cranial Nerves & Spinal Nerves.
44. Effect of two parts of Autonomic nervous system (ANS) Comparison between Sympathetic and Parasympathetic nervous system. (Important question for 2026)
45. Concept of Reflex Action and types of reflexes with examples. (Important question for 2026)
46. Pathway of Reflex Action. (Important question for ICSE 2026)
47. General study of the following endocrine gland with special reference to Adrenal, Thyroid, Pancreas, Pituitary & Gonads. (Important question for 2026)
48. Diagram based questions on Adrenal, Thyroid, Pituitary and Pancreas. (Important question for ICSE 2026)
49. General properties of Hormones and Differences between enzymes & hormones, Endocrine & Exocrine glands along with examples. (Important question for 2026)
50. Structure & function of different parts of Male & Female Urinogenital System. (Important question for ICSE 2026)
51. Course of sperm flow & Parturition of the baby (Only flow sheet diagram). (Important question for ICSE 2026)
52. Different phases of Menstruation (only names are required & days) (Important question for ICSE 2026)
53. Concept and Definition of Menarche, Menopause, Gestation, Implantation, Different Extra embryonic membrane & Placenta. Functions of Placenta and Amniotic fluid (Important question for ICSE 2026)
54. Different terminologies related with Population (Important question for ICSE 2026)
55. Definition and Concept of Tubectomy, Vasectomy & IUCDs (Lippe's Loop & Copper-T). (Important question for ICSE 2026)
56. Population Explosion- Definition, Factors, Consequences & Remedial measures (Important question for 2026)
57. Entire chapter of Evolution with special reference to the comparative chart on Human Evolution (Important question for 2026)
58. Different types of Environmental Pollution ---- (Sources, Effects & Preventions) a tabular format given in the class (Important question for ICSE 2026)
59. Concept of Biodegradable and Non-Biodegradable Pollutants with examples. (Important question for ICSE 2026)
60. Global Issues on Environment----- Acid rain, Green House Effect, Global Warming, Ozone depletion, Eutrophication (their major causes, effects & Preventive measures). (Important question for ICSE 2026)
61. Concept about Smog, BOD , DDT & ODF along with its expanded forms. (Important question for ICSE 2026)
62. Minamata & Itai-itai disease – causes & effects.
63. Fertilizers & Manures.
64. Concept about Biofertilizers.
65. Vehicular Standards for the emission of smoke from automobiles / Euro II/ Bharat Stage IV & VI.
66. Swachh Bharat Abhiyan its movement, year of initialization and its objectives. (Important question for ICSE 2026)

FULL FORMS

1.	DNA	-	Deoxyribonucleic acid
2.	RNA	-	Ribonucleic acid
3.	ATP	-	Adenosinetri phosphate
4.	NADP	-	Nicotinamide adenine dinucleotide phosphate
5.	IAA	-	Indole acetic acid
6.	IBA	-	Indole butyric acid
7.	2,4, -D	-	Dichlorophenoxy acetic acid
8.	NAA	-	Naphthalene acetic acid
9.	ABA	-	Abscisic acid
10.	GA	-	Gibberellic acid
11.	SAN	-	Sino-atrial node
12.	AVN	-	Atrioventricular node
13.	Hb	-	Heamoglobin
14.	Rh	-	Rhesus antigen
15.	ECG	-	Electrocardio gram
16.	CNS	-	Central nervous system
17.	ANS	-	Autonomic nervous system
18.	PNS	-	Peripheral nervous system
19.	FSH	-	Follicle stimulating hormone
20.	LH	-	Luteinising hormone
21.	ACTH	-	Adrenocortico tropin Hormone
22.	TRH	-	Thyroid realizing hormone
23.	IUDS	-	Intra uterine devices
24.	MTP	-	Medical termination of pregnancy
25.	ZPG	-	Zero population growth
26.	AIDS	-	Aquired immuno deficiency syndrome
27.	BOD	-	Biological oxygen demand
28.	dB	-	Decibles
29.	CNG	-	Compressed natural gas
30.	DDT	-	Dichloro – diphenyl trichloro ethane
31.	CFCs	-	Chloro Fluoro carbons
32.	O3	-	Ozone
33.	ODF	-	Open Defecation free
34.	TSH	-	Thyroid stimulating hormone (Protein)
35.	MSH	-	Melanocyte Stimulating Hormone(Peptide)
36.	GH	-	Growth Hormone
37.	EEG	-	Electro encephalogram

Name: _____

SCIENTIFIC NAMES

1. Pea - Pisum sativum
2. Oat plant - Avena sativa
3. Man - Homo sapiens
4. Peppered moth - Biston betularia
5. Dark coloured moth - Biston betularia carbonaria

SCIENTISTS NAMES

1. **HOFMEISTER** discovered chromosomes for the first time
2. **FREDERICK MIESHER** were first discovered the two nuclei acids DNA and RNA
3. Chromosomes in animals were first studied in 1882 by a german scientist **WALTHER FLEMMING**
4. The shape of the DNA molecule was studied by **ROSALIND FRANKLIN** in 1953.
5. The structure of DNA molecule was studied by **WATSON** and **CRICK** in 1953
6. **GREGOR MENDEL** father of genetics.
7. The term hormone was given by **WILLIAM BAYLISS** and **EARNEST STARLING**
8. The first indication of plant growth regulator was given by **DARWIN** in 1880
9. Auxin first discovered by **F.W.WENT**
10. Gibberellins were first reported in japan by **E.KUROSAWA** in 1926
11. **CARL LANDSTEINER** 1990 discovered the blood groups.
12. Rh factor –Rhesus monkey discovered by **LANDSTEINER** and **WEINER** in 1940
13. **THOMAS EDISON** is called as father of endocrinology.
14. **SECRETIN** was the first hormone to be discovered by **WILLIAM M BAYLISS** and **ERNEST H STARLING**
15. Dryopithicus (These arms and legs were almost equal in length) It was discovered by **Edouard Lartet in France.**
16. Ramapithecus was discovered by **Edward Lewis.**
17. Australopithecus was discovered by **Ramond Dart.**
18. Homo habilis (The Handy man) was discovered by **L.S.B. Leakey** and his wife **Mary Leakey.**
19. Homo erectus (The upright or erect man) was discovered by **Mayer.**
20. Neanderthal man was discovered by **C Fahlrott.**
21. Cro – Magnon Man (Homo sapiens fossillis) were discovered by **Mac Gregor.**
22. Lamarck's theory of inheritance of acquired characters (**Lamarckism**)
23. Darwin's theory of Natural selection (**Darwinism**)
24. Father of Evolution "**Darwin**".

Prepared by: Mr. Sumit Roy [Senior Biology Teacher] (9830431393 / 7044929409 / 9874577160)
M.Sc (Zoology), Gold Medalist of Vidyasagar University, B.Ed . Post Graduate Diploma in Environmental Biotechnology (C.U.)

E mail: - sumitroy888@gmail.com

sumit@ssa.org.in

sumitroy98304@gmail.com